



S V V SAI SRI VALLABH PATANENI

Control Strategy Engineer · Global Graduate Program 2026

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PROFILE

My work has followed one thread: making energy systems behave the way they should through control. It started at Ecogenium e.V., where I wrote and validated real-time energy-management and temperature-control strategies in Python and Simulink that lifted our vehicle's efficiency by ~80%. At Robert Bosch GmbH I took this to industrial level, developing rule-based and predictive power-split control strategies, calibrating them with Genetic-Algorithm and Gradient-Descent optimisation, and benchmarking against Dynamic Programming and ECMS. At DLR I am now building a fuel cell system controller in MATLAB and C++ for direct deployment to hardware, with start-up, purge, subsystem and safety state logic. I am applying to Goldwind's Global Graduate Program 2026 to bring this control-strategy and model-based-development experience to your turbine control team in Frankfurt.

PROFESSIONAL EXPERIENCE

Operating Strategy / System Simulation Intern 09/2025 – 12/2025
Robert Bosch GmbH Schwieberdingen, Germany

- Control Strategy Development:** Engineered a new MATLAB control strategy for Strong Fuel Cell Hybrid EVs that manages dynamic loads, thermal stress, low-voltage operation and SOC swing, parameterised via MAPs for different operating conditions and duty cycles.
- Model-Based Development & Validation:** Developed and validated an iterative rule-based power-split strategy in the Simulink (FC)EVsim module, correlating simulation outputs against reference measurement data to ensure high model fidelity.
- Optimisation & Benchmarking:** Executed multi-objective, multi-variable parametric optimisation using Genetic Algorithms and Gradient Descent to generate control MAPs, benchmarked against Dynamic Programming and ECMS standards.
- Predictive Control & Publication:** Researching predictive control extensions using live GPS and road-elevation data to anticipate power demand; co-authoring a peer-reviewed paper on the control strategy.

Werkstudent, Fuel Cell System Control (Aircraft) 01/2026 – Present
DLR, Innovation Center for Small Aircraft Technologies (INK) Wuerselen (Aachen), Germany

- Control-Oriented Modelling:** Built a MATLAB/Simulink digital twin of the aircraft fuel cell thermal management system to derive control specifications (volumetric flow, coolant temperature range, coolant pressure) for the ~150 kW system.
- Controller Development (Thesis):** Developing the fuel cell system control (start-up, purge control, cathode and anode subsystem control, temperature control, safety protocols) in MATLAB and C++ for direct deployment to the physical controller.
- System Integration:** Selecting subsystem components against derived specifications and designing the system rig in CATIA V5 for integration and test.

Control & Energy-Management Lead (Team Lead) 06/2024 – Present
Ecogenium e.V (RWTH student team), Shell Eco-Marathon Aachen, Germany

- Real-Time Control Strategy:** Architected and deployed energy-management and fuel-cell temperature-control strategies in Python/Simulink, validated with Model-in-the-Loop simulation, boosting on-track efficiency by ~80% and securing a runner-up finish at Shell Eco-Marathon 2025.
- Control Validation on Hardware:** Engineered and programmed a standalone test bench and correlated control models against real measurement data (polarisation curves, purging, thermal sweeps) to validate strategy before deployment.
- Leadership:** Led a multidisciplinary team building Germany's only student-built hydrogen fuel cell vehicle from concept to competition, coordinating control, integration and on-track troubleshooting under deadlines.

Autonomous Systems Engineer (Academic Project) 04/2025 – 07/2025
Institute for Automotive Engineering (IKA), RWTH Aachen Aachen, Germany

- Controller Design & Validation:** Developed models and tuned PID and MPC-based controllers in MATLAB/Simulink and Python for Adaptive Cruise Control and Lane Keep Assist, validated through real-time testing on a mock vehicle and in CARLA against real-world results.

KEY SKILLS

CONTROL METHODS	PID control · Model Predictive Control (MPC) · rule-based control · predictive control · feedforward / feedback control · supervisory & state-machine control · gain scheduling · real-time control
STRATEGY & CALIBRATION	control strategy design · operating & energy-management strategy · control MAP / lookup-table calibration · multi-objective optimisation · Genetic Algorithms · Gradient Descent · DP / ECMS benchmarking · KPI analysis
MODEL-BASED DEVELOPMENT	MATLAB/Simulink · digital twin · Model-in-the-Loop (MIL) · model & measurement validation · C++ deployment to controller · CARLA
PROGRAMMING & DATA	MATLAB · C/C++ · Python (NumPy, Pandas, scikit-learn, PyTorch) · signal processing · Power BI
TOOLS	Siemens Teamcenter · Git · Tableau

EDUCATION

M.Sc. Automotive Engineering 09/2023 – 06/2026
RWTH Aachen University GPA 1.7 (1.2 core courses)
Modules: Simulation Sciences, Mechatronics, Machine Learning / Data Science, Alternative & Mobile Propulsion Systems. Thesis: fuel cell system control (MATLAB/C++).

B.Tech Mechanical Engineering 06/2019 – 04/2023
Jawaharlal Nehru Technological University, Hyderabad GPA 1.3 (German scale)

SELECTED PROJECTS & PUBLICATIONS

- Co-author (in preparation): rule-based powertrain control strategy with optimisation and benchmarking, based on the Bosch internship.
- Vehicle Sensor Data Analysis (IKA-RWTH): Python pipelines for camera/LiDAR/radar, CNN traffic-sign recognition, Power BI dashboards.
- Paper: A Comprehensive Analysis of Two Innovative Aircraft Design Configurations (IJSED).

LANGUAGES

English (C1, fluent) · German (B1, intermediate) · Telugu / Hindi (native)